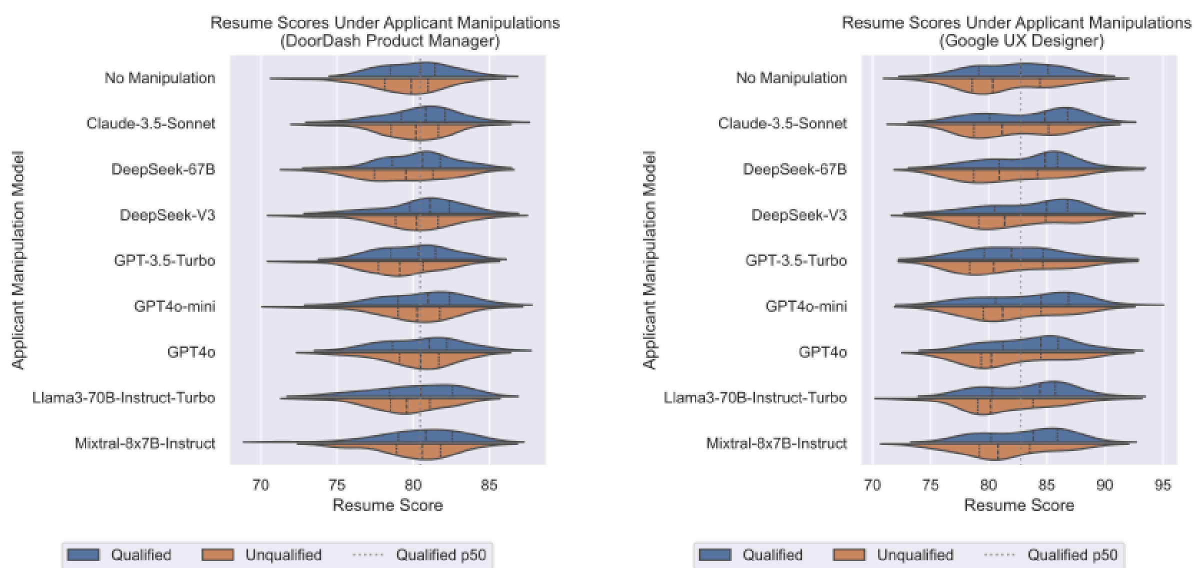
[Literature Review] Two Tickets are Better than One: Fair and Accurate Hiring Under Strategic LLM Manipulations



(a) DoorDash Product Manager Job Posting (b) Google UX Designer Job Posting

The paper titled "Two Tickets are Better than One: Fair and Accurate Hiring Under Strategic LLM Manipulations" by Lee Cohen, Jack Hsieh, Connie Hong, and Judy Hanwen Shen addresses the implications of generative AI tools on the hiring process, particularly in relation to fairness and strategic manipulations by candidates.

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disparities in access to these tools. The primary problem highlighted is that candidates with better access to more capable LLMs may gain an unfair advantage in the hiring process, resulting in skewed hiring outcomes based on the manipulation of application materials rather than genuine qualifications.

Methodology Overview:

The core methodology introduced is an adaptation of the strategic classification framework that accounts for stochastic manipulations enabled by LLMs. The authors propose a "two-ticket" scheme, where the hiring algorithm evaluates both the original resume and a manipulated version refined through an LLM, in order to counteract disparities related to unequal access to generative AI.

Detailed Methodology Description:

1. **Strategic Classification Framework**: The problem is framed within strategic classification, where candidates manipulate their resumes to influence classifier outcomes. Candidates aim to maximize their chances of selection while minimizing the costs of manipulation. The authors argue that traditional classifiers may be vulnerable to manipulation, as unqualified candidates may present resumes that appear qualified through strategic modifications.

2. **Two-Ticket Scheme**:

   - In this scheme, the hiring algorithm generates a second version of each submitted resume using the employer's own LLM. This means that both the original and manipulated resumes are assessed to make a hiring decision.
   - The authors claim that using a second, potentially superior LLM manipulation reduces disparate outcomes by providing candidates

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- The mathematical formulation defines LLM manipulation as a random function transforming the resume features stochastically, preserving core features while modifying stylistic ones.

3. **Theoretical Guarantees**:

- The authors prove that their two-ticket approach enhances both fairness and accuracy. They define a "No False Positives" objective, where the goal is to maximize the true positive rate (TPR) while ensuring no false positives occur.
- The formulation shows that increasing the number of "tickets" (i.e., manipulations) progressively reduces bias, as it approaches a state where the outcomes become independent of group membership.

4. **Empirical Validation**:

- The methodology is validated through a case study utilizing real resumes screened by an open-source resume scoring system. The authors sampled resumes and manipulated them using various LLMs, recording the relevance scores to assess the performance of the proposed two-ticket scheme.
- Results indicate that the two-ticket scheme improved true positive rates (TPR) and reduced TPR disparities, especially in conditions where unprivileged candidates used weaker models.

Conclusion:

The theoretical contributions emphasize the necessity of incorporating multi-level evaluations within hiring algorithms to mitigate biases introduced through strategic manipulations. The empirical findings corroborate the theoretical analyses, suggesting that the two-ticket scheme can indeed enhance both fairness and accuracy in the hiring process. The authors also acknowledge that

Future Directions:

The authors suggest that future research could expand beyond hiring settings to explore other scenarios where candidates manipulate their submission materials. They note the importance of refining the model to account for variations in LLM performance and respond to dynamic changes in the provider landscape of generative AI tools. Furthermore, exploring the use of human evaluators alongside algorithmic assessments in real-world settings could yield important insights into the efficacy of the two-ticket scheme and similar methodologies.

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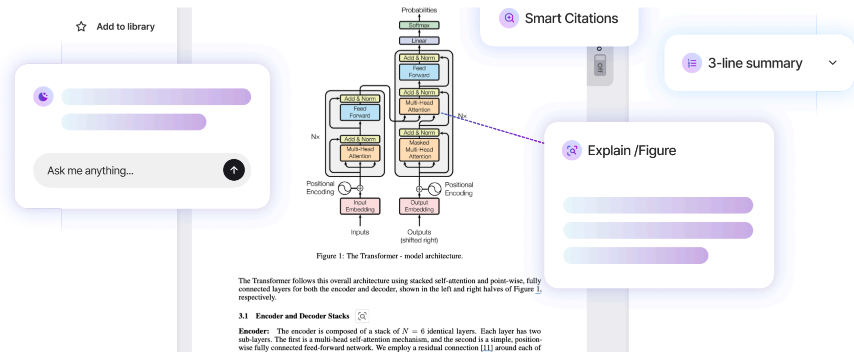

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